

Google Workspace Deployment & Security Implementation (Live Environment)

Project Overview

This project demonstrates the deployment and security configuration of a Google Workspace environment for a small business.

The implementation focuses on identity management, email security, administrative controls, collaboration tools, and operational workflows. It reflects real-world IT administration tasks while accounting for licensing and production environment constraints.

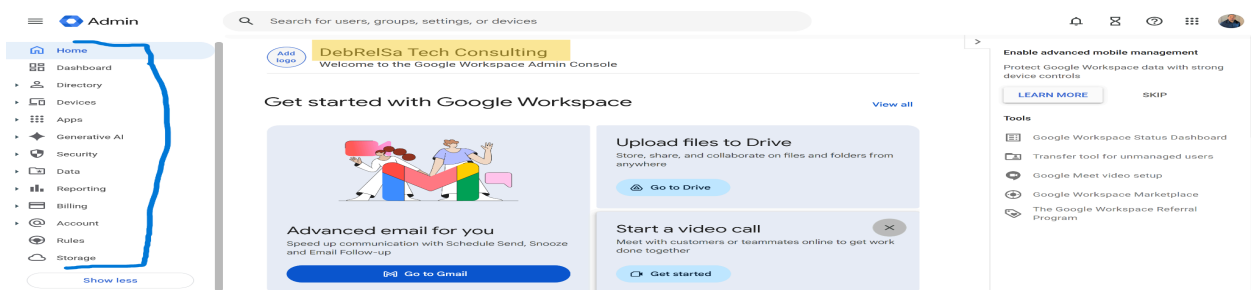
Phase 1 — Environment Validation & Administrative Access

Administrative access was verified using a Super Admin account. Core services (Directory, Security, Apps, Reporting, Billing) were accessible and properly configured.

The primary domain was validated and Gmail functionality was confirmed through DNS (MX records) and a successful login test.

This phase established a working administrative baseline.

Figure 1: Shows full admin console interface



Phase 2 — DNS Configuration & Email Authentication

Email routing was verified through correct MX records.

Missing SPF was identified and remediated to prevent spoofing. DKIM was configured for message integrity, and DMARC was implemented for policy enforcement and reporting.

All email authentication records were successfully validated.

Figure 2: Domain Verification

Manage domains Add a domain Change primary domain			
Domain	Type	Status	Actions
debrlsatechconsulting.com	Primary Domain ?	Verified Gmail activated	Set up redirect Add Users
debrlsatechconsulting.com.guest.google	Guest Domain ?	Verified	
debrlsatechconsulting.com.test-google-a.com	Test domain alias ?	Active	Deactivate

Figure 3: Configured Google Workspace MX records

Google Workspace				
TYPE ?	NAME ?	PRIORITY ?	TTL ?	DATA ?
MX	@	1	4 hrs	aspmx.l.google.com
MX	@	5	4 hrs	alt1.aspmx.l.google.com
MX	@	5	4 hrs	alt2.aspmx.l.google.com
MX	@	10	4 hrs	alt3.aspmx.l.google.com
MX	@	10	4 hrs	alt4.aspmx.l.google.com

Figure 4: Added missing SPF record

Custom records Custom records for your domain. ADD RECORD				
TYPE ?	NAME ?	PRIORITY ?	TTL ?	DATA ?
TXT	@	N/A	4 hrs	google-site-verification=s47U8nzKX38EJUY...
TXT	@	N/A	4 hrs	v=spf1 include:_spf.google.com ~all

Figure 5: Added DKIM record

Custom records Custom records for your domain. ADD RECORD				
TYPE ?	NAME ?	PRIORITY ?	TTL ?	DATA ?
TXT	@	N/A	4 hrs	google-site-verification=s47U8nzKX38EJUY...
TXT	@	N/A	4 hrs	v=spf1 include:_spf.google.com ~all
TXT	google._domainkey	N/A	4 hrs	v=DKIM1; k=rsa; p=MlIBijANBgkqhkiG9w0B...

Figure 6: Added DMARC record

Custom records

Custom records for your domain.

ADD RECORD

TYPE ?	NAME ?	PRIORITY ?	TTL ?	DATA ?
TXT	google_domainkey	N/A	4 hrs	v=DKIM1; k=rsa; p=MIIBIJANBgkqhkiG9w0B...
TXT	@	N/A	4 hrs	google-site-verification=s47U8nzKX38EJUY...
TXT	@	N/A	4 hrs	v=spf1 include:_spf.google.c...
TXT	_dmarc	N/A	4 hrs	v=DMARC1; p=quarantine; rua=mailto:you@...

Phase 3 — Organizational Structure Design

Organizational Units (OUs) were created to segment users by function (Executive, IT, Marketing, Operations, Contractors).

This structure enables scalable policy enforcement and aligns with real-world enterprise design.

Figure 7: OU's not populated

Organizational units

Manage organizational units Showing 2 organizational units		Create organizational unit
Search for organizational units		
Name	Description	
DebRelSa Tech Consulting	DebRelSa Tech Consulting	
Workspace Guests	A dedicated Org Unit for external guests	

Figure 8: OU's populated

Manage organizational units Showing 2 organizational units		Create organizational unit
Search for organizational units		
Name	Description	
DebRelSa Tech Consulting	DebRelSa Tech Consulting	
Workspace Guests	A dedicated Org Unit for external guests	
Contractors	Restricted external users	
Executive	Leadership / high-security users	
IT	Admins / technical users	
Marketing	Content / external tools	
Operations	Day-to-day business users	

Phase 4 — User Provisioning (Constraint-Based Simulation)

User provisioning was reviewed but not executed due to licensing and production constraints.

The process for creating users and assigning OUs was validated, reflecting real-world change management practices.

Figure 9: Users Created List

Manage organizational units Showing 2 organizational units		Create organizational unit
Search for organizational units		
Name	Description	
▼ DebRelSa Tech Consulting	DebRelSa Tech Consulting	
📁 Workspace Guests	A dedicated Org Unit for external guests	
Contractors	Restricted external users	
Executive	Leadership / high-security users	
IT	Admins / technical users	
Marketing	Content / external tools	
Operations	Day-to-day business users	

Phase 5 — Group Configuration & Access Design

Google Groups were designed for communication and access control.

Although users were not added, group structure and intended usage were documented to simulate a full deployment model.

Figure 10: Changing user's OU

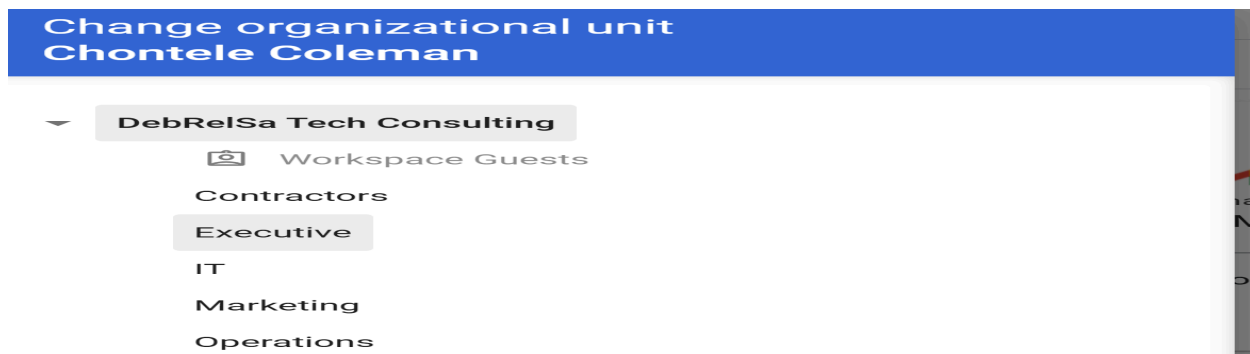
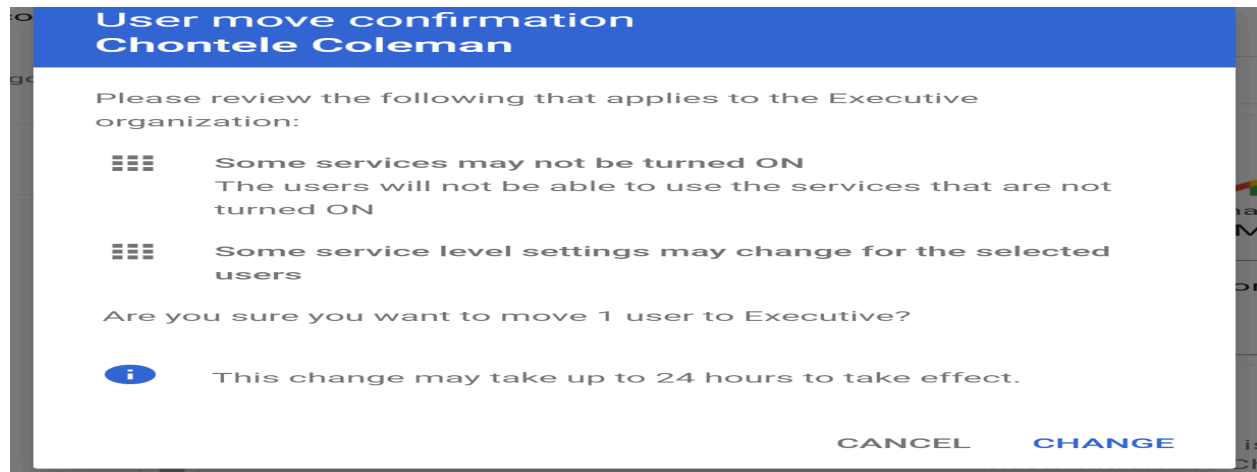


Figure 11: Warning message if OU is changed



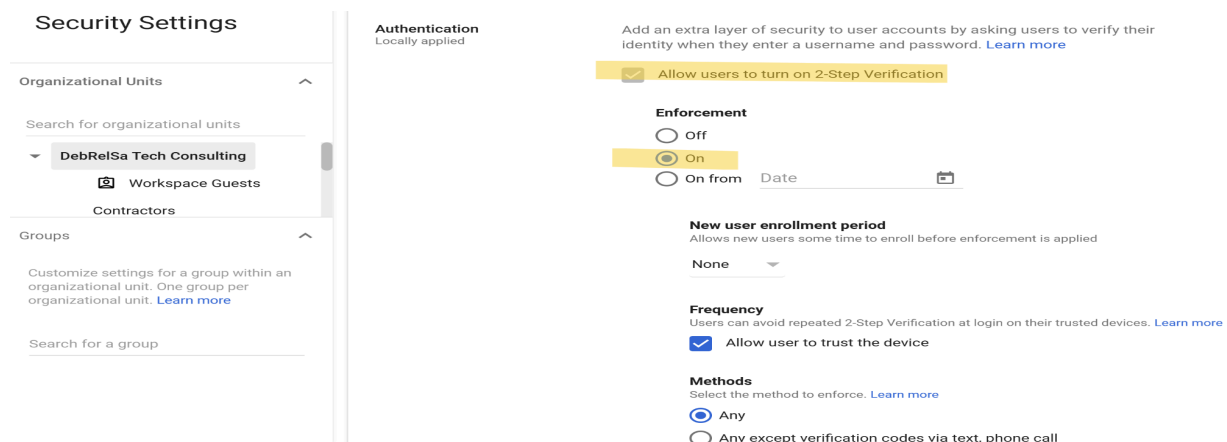
Phase 6 — Security Controls & Hardening

Two-Step Verification (2SV) was enabled for administrative access.

Security settings and alerts were reviewed, with noted limitations due to the Business Starter tier.

Core services (Gmail, Drive, Calendar, Chat) were verified as active.

Figure 12: Two-Step Verification turned on



Phase 7 — Collaboration Environment Configuration

Shared Drives, calendars, and Chat spaces were created to simulate business collaboration.

Access was designed using group-based permissions aligned with organizational roles.

Figure 13: Shared Drives created

Shared drives

Hidden shared drives

Q Shared drive name

Name	Members
 Company Files	1 person
 HR	1 person
 IT Documentation	1 person
 Marketing Assets	1 person

Figure 13: Members Managed

 Company Files
1 person

Type | People | Modified | Source

Manage members

Add people and groups

 Chontele Coleman (you)
chontele@debrelsattechconsulting.com

Manager

Done

Figure 14: Created company calendar

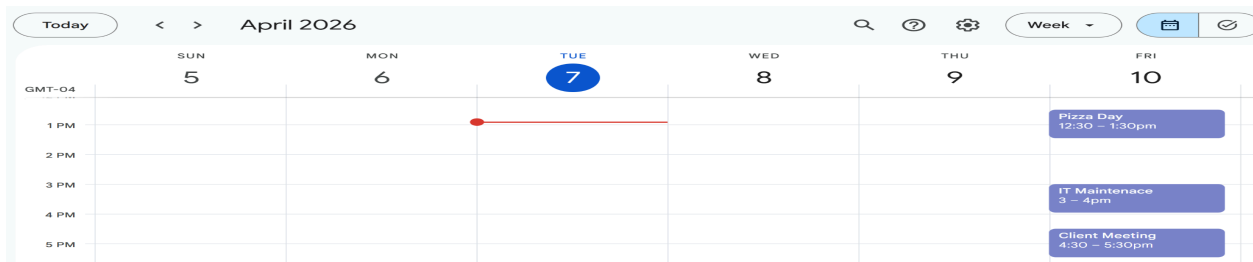
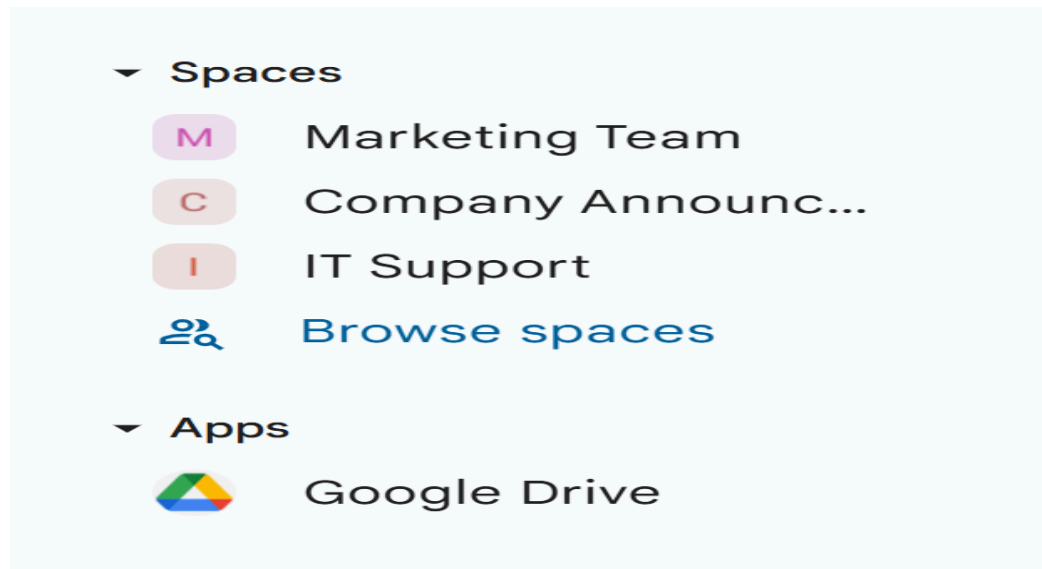


Figure 15: Chat spaces created



Phase 8 — Onboarding Workflow Development

A standardized onboarding process was developed, including:

- Account creation
- OU assignment
- Group membership
- Security enforcement

Supporting documents (SOP and checklist) are maintained separately.

Phase 9 — Offboarding Workflow Development

A secure offboarding process was created, including:

- Account suspension
- Access removal
- Data ownership transfer

A checklist ensures consistent execution and prevents residual access.

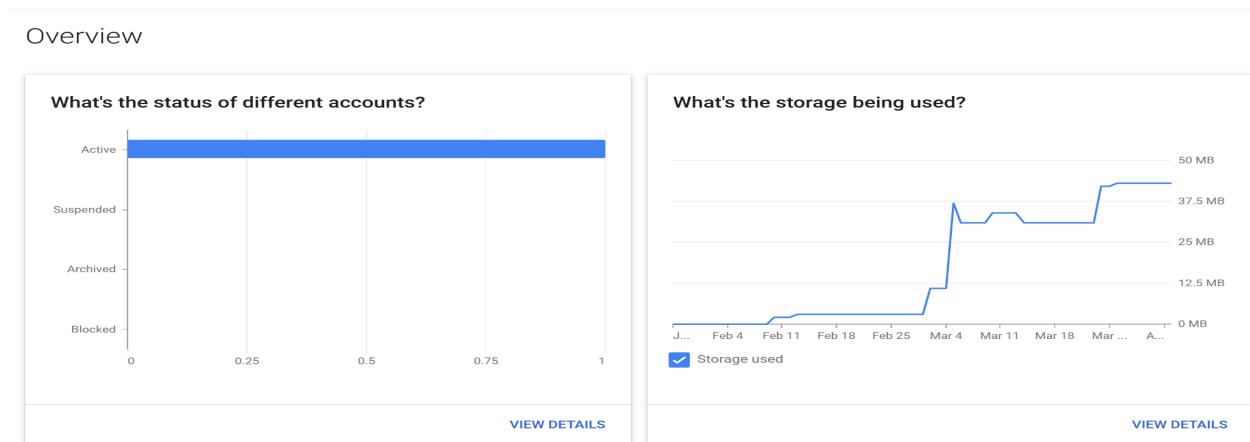
Phase 10 — Reporting & Monitoring

Monitoring capabilities were reviewed, including admin logs, login activity, and 2SV tracking.

Advanced features were noted as limited due to licensing.

Supporting monitoring documentation is maintained separately.

Figure 16: Reporting Dashboard



Conclusion

This project demonstrates the deployment and security configuration of a Google Workspace environment within a live business setting. All work was performed in a production environment, including the organization's Google Workspace tenant and Squarespace-managed DNS.

The implementation covered identity management, email authentication (SPF, DKIM, DMARC), access control, and security measures such as multi-factor authentication, along with supporting onboarding, offboarding, and monitoring processes.

This work reflects practical, real-world experience managing business systems while adapting to operational constraints and maintaining security best practices.